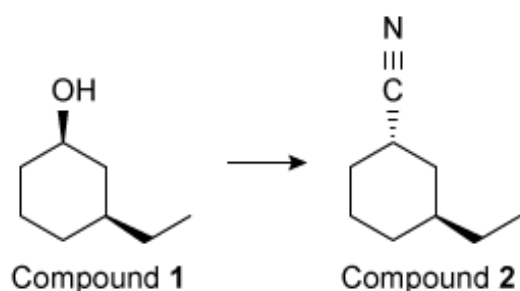


Consider the reaction below.



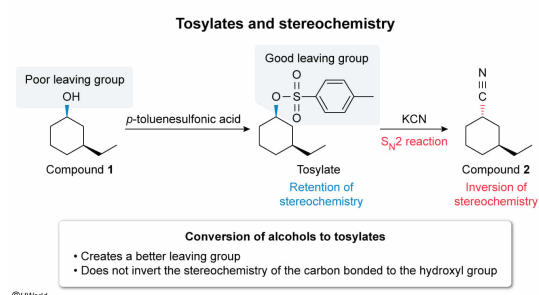
Compound 2 is prepared from Compound 1 by:

- ☐ A. an S_N2 reaction with KCN. [29%]
- ☐ B. oxidation with PCC, followed by nucleophilic addition of KCN and an acidic workup. [24%]
- ☐ C. an S_N2 reaction with PBr_3 , followed by an S_N2 reaction with KCN. [21%]
- ✓ ☒ D. condensation with *p*-toluenesulfonic acid, followed by an S_N2 reaction with KCN. [25%]

Correct

25% answered correctly

Explanation:



Substitution reactions involve the reaction of **nucleophiles with electrophiles** and require a good **leaving group**. Leaving groups accept a pair of electrons and are often stabilized by **resonance** and/or an **inductive effect**. In a substitution reaction, the nucleophile attacks an electrophile and displaces the leaving group—a process that results in bond formation between the nucleophile and electrophile and bond cleavage between the electrophile and leaving group.

Hydroxide is a poor leaving group, and the carbon bonded to the hydroxyl group is a weak electrophile. Therefore, a hydroxyl group is unfavorable as a leaving group in a substitution reaction unless a modification is made to make it a better leaving group. Conversion to a **mesylate** or **tosylate** **improves** the **hydroxyl group's leaving ability**, as **mesylate and tosylate anions** are stabilized by resonance. Conversion of **chiral** alcohols to mesylates or tosylates **does not change** the **stereochemistry** of the carbon bonded to the hydroxyl group.

In this question, the hydroxyl group in Compound 1 is converted to a **nitrile** with **opposite** stereochemistry, suggesting an **S_N2 reaction** is involved. Because hydroxyl groups are poor

leaving groups, it must first be modified. **Condensation** of Compound **1** with **p-toluenesulfonic acid** produces a tosylate (a better leaving group) while *retaining* the stereochemistry of the carbon bonded to the tosylate. An **S_N2 reaction with KCN** then causes inversion of the stereochemistry and produces Compound **2**.

(Choice A) Because alcohols are poor leaving groups, Compound **1** reacting with KCN via an S_N2 reaction is *unfavorable*.

(Choice B) **Oxidation** of the secondary alcohol in Compound **1** produces a ketone. Nucleophilic addition of KCN to the ketone followed by an acidic workup produces a **cyanohydrin**, *not* a nitrile.

(Choice C) S_N2 reaction with PBr₃ generates an alkyl bromide and *inverts* the stereochemistry at the alcohol chiral carbon. The subsequent S_N2 reaction with KCN *also* inverts the stereochemistry at the chiral carbon bonded to the bromide, resulting in **net retention of stereochemistry**.

Educational objective:

Although a hydroxyl group is a poor leaving group, conversion to a mesylate or tosylate makes it an improved leaving group. Conversion of chiral alcohols to mesylates or tosylates does not change the stereochemistry of the carbon bonded to the hydroxyl group.

Time spent: 0 sec

Subject:
Organic Chemistry

QID: 404171

System:
Functional Groups and Their Reactions